

CSE4067

Introduction to Blockchain Programming

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Zero-Knowledge Proof

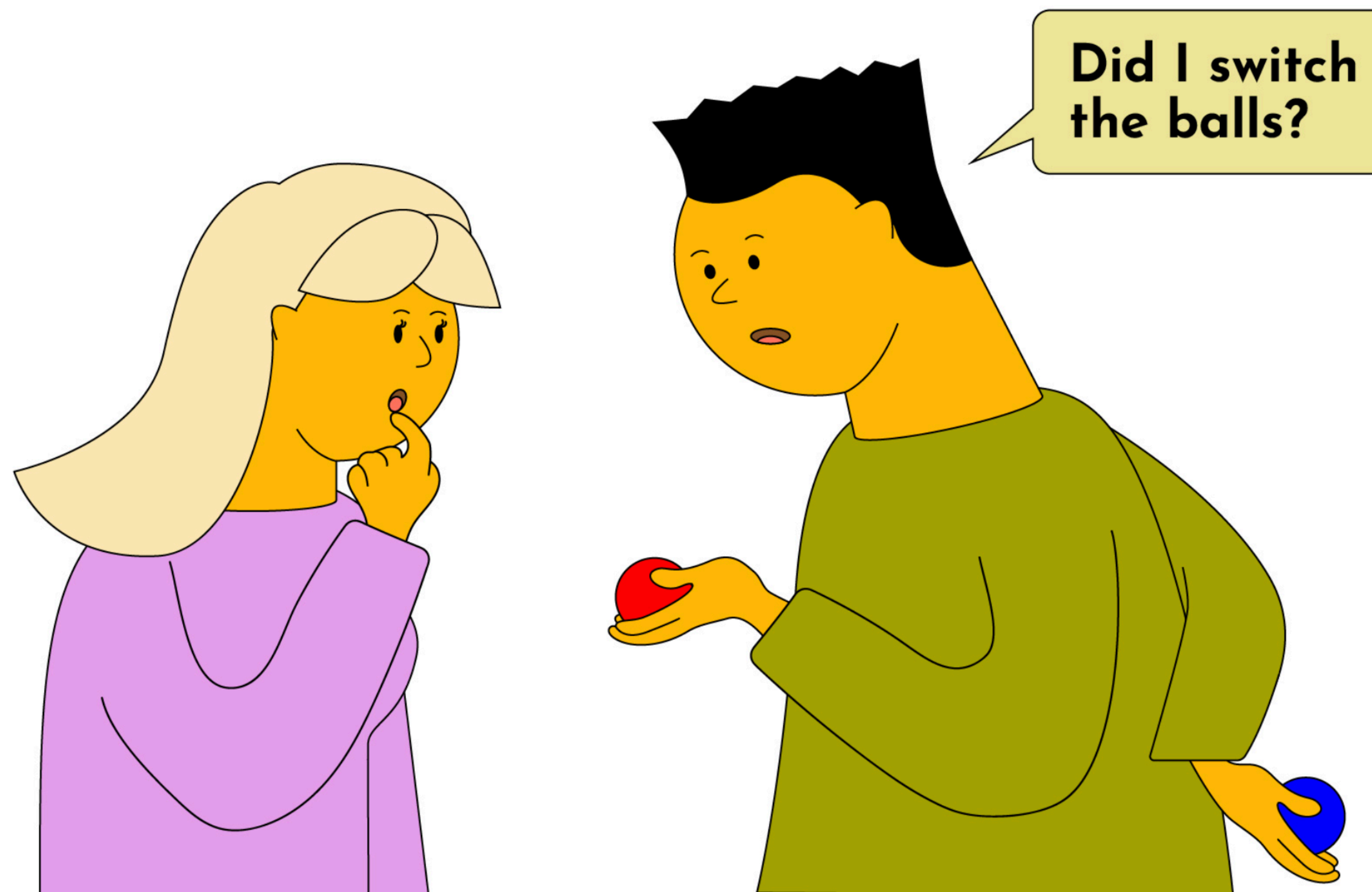
A zero-knowledge proof is a method that allows one party (Peggy) to prove to another (Victor) that she knows a secret value that fulfills some constraints — without revealing any additional information to Victor, only that she knows this secret value.

- **Prover:** The prover (“Peggy”) wants to prove to the verifier (“Victor”) that she knows the witness without revealing the witness.
- **Verifier:** The verifier (“Victor”) needs to verify that the prover does know the witness without revealing the witness to himself or anyone else.

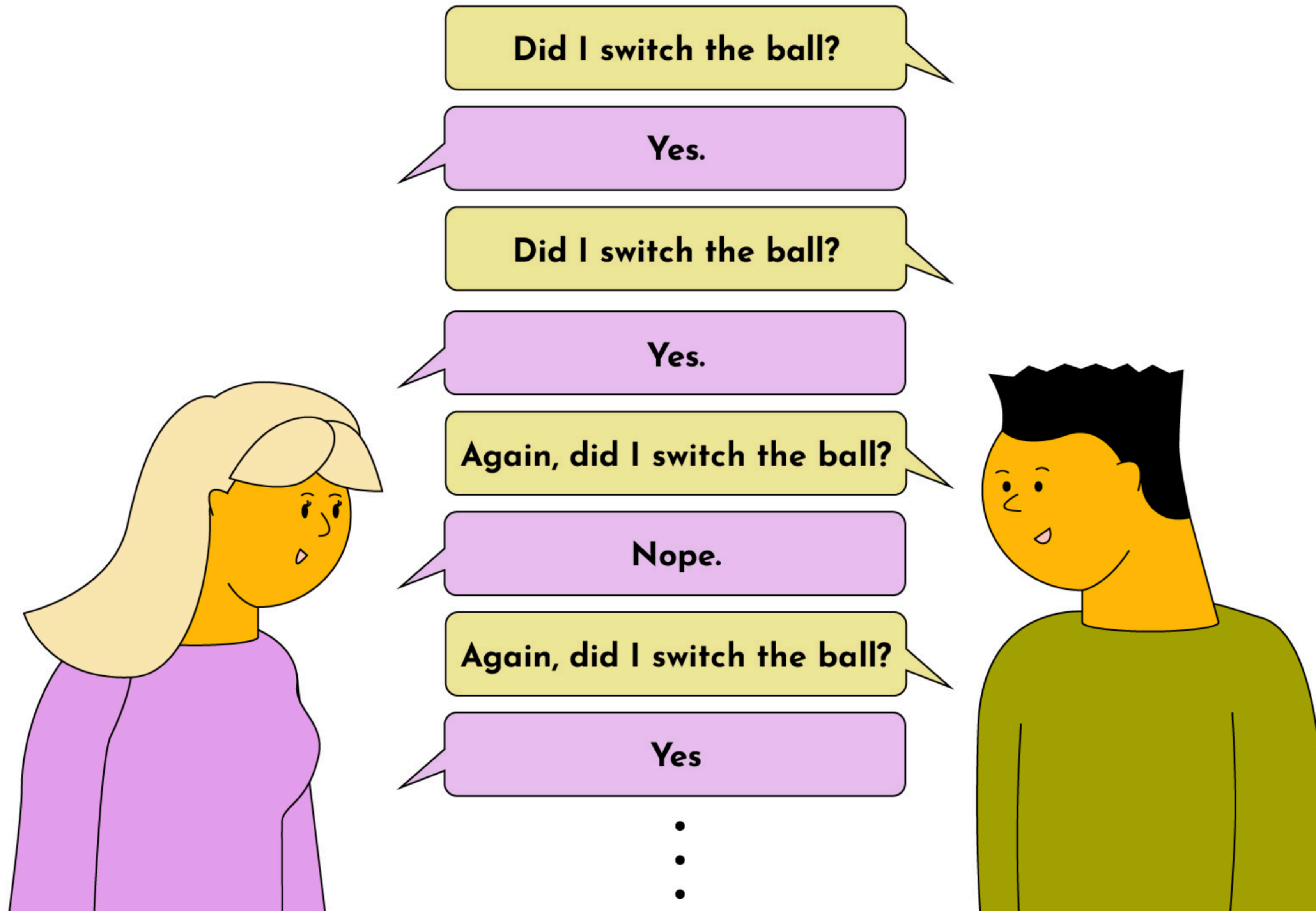


Zero-Knowledge Proof: Color Blindness

Imagine Peggy has two colored balls — one is red, and the other is blue. Since Victor is colorblind, he might not believe that the balls are different colors. Peggy, who isn't colorblind, wants to prove to Victor that the balls are different colors, even though she can't actually reveal the colors.

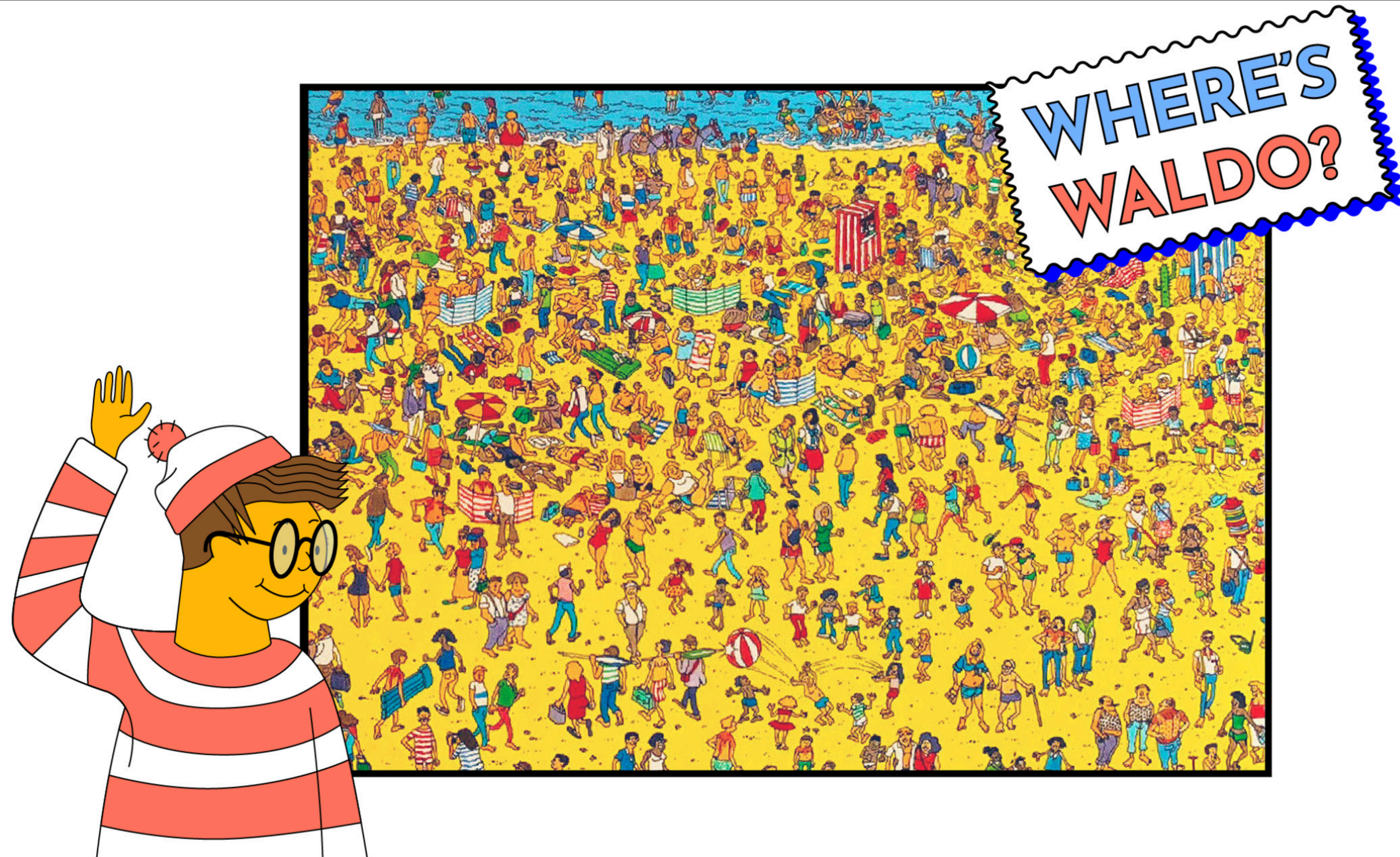


Zero-Knowledge Proof: Color Blindness



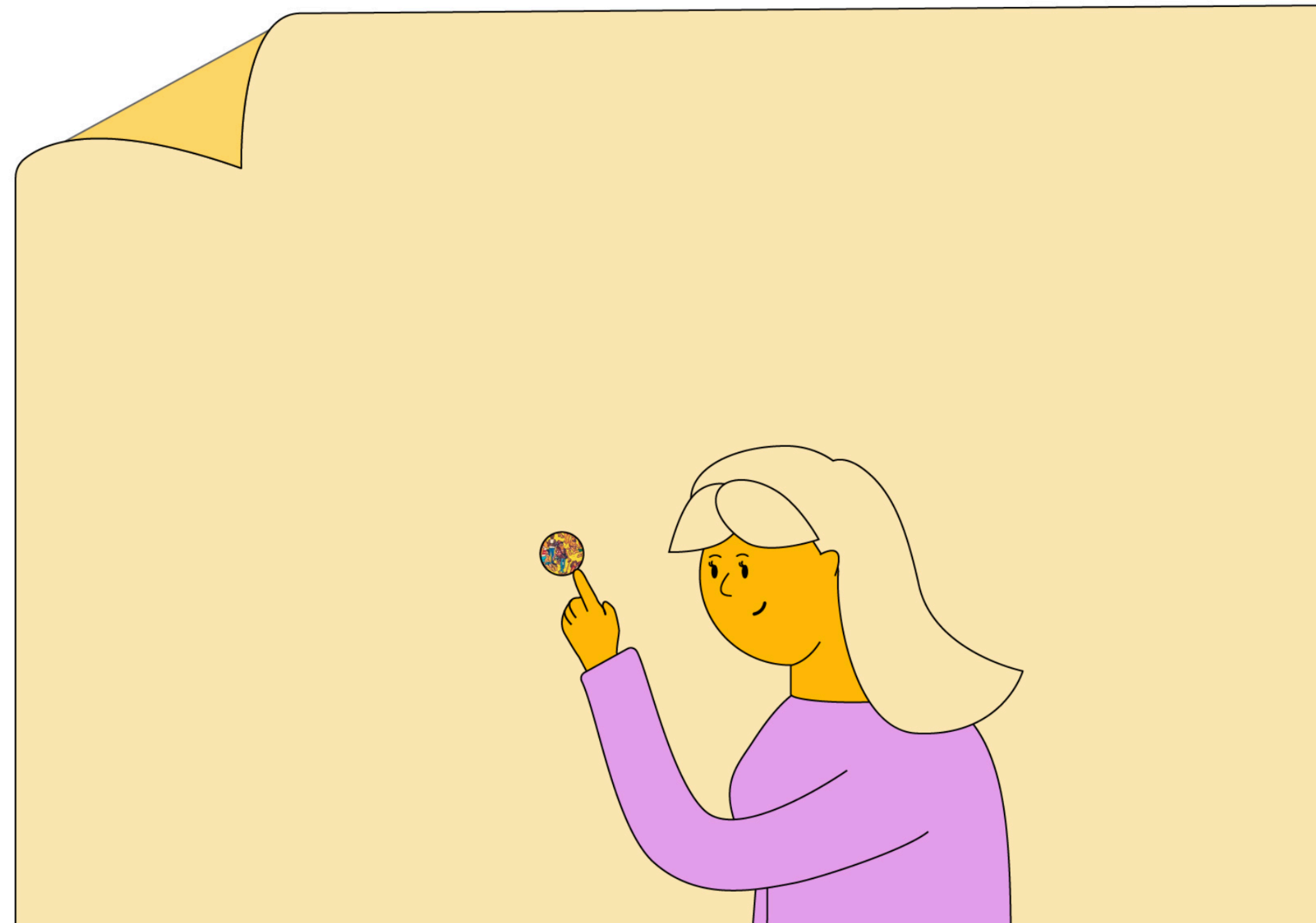
Zero-Knowledge Proof: Waldo

“Where’s Waldo?” is a game where you have to find the titular Waldo (who looks like the image below) in a sea of characters and shapes that look like him.

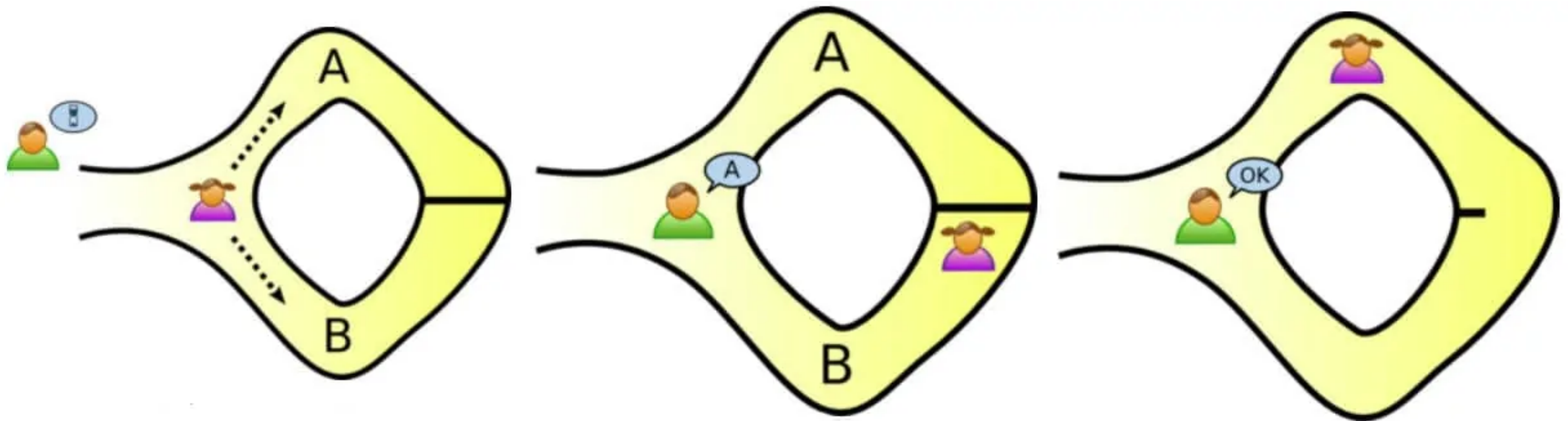


Zero-Knowledge Proof: Waldo

Imagine Peggy takes a very large and opaque piece of paper and cuts a small hole in it. She places the hole on top of the puzzle such that Waldo is visible through the hole. This shows that she knows where Waldo is, but Victor won't be able to figure out Waldo's location within the puzzle. You could imagine that the piece of paper is much larger than the original puzzle so figuring out the location of Waldo based on the cutout is not possible.



Zero-Knowledge Proof: Ali Baba Cave

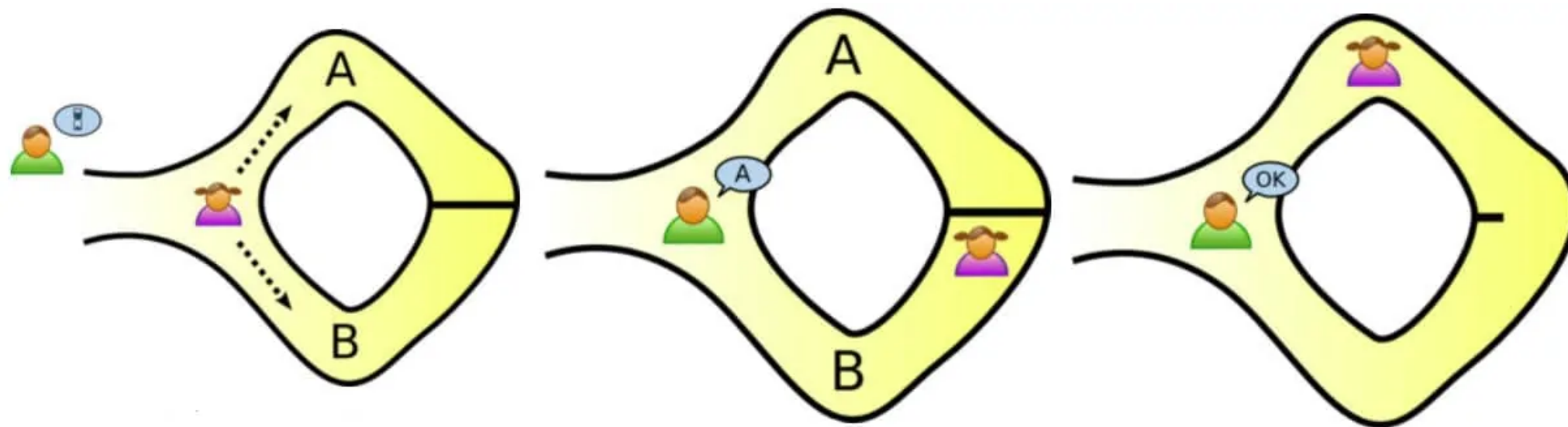


Zero-Knowledge Proof: Ali Baba Cave

How to Explain Zero-Knowledge Protocols to Your Children

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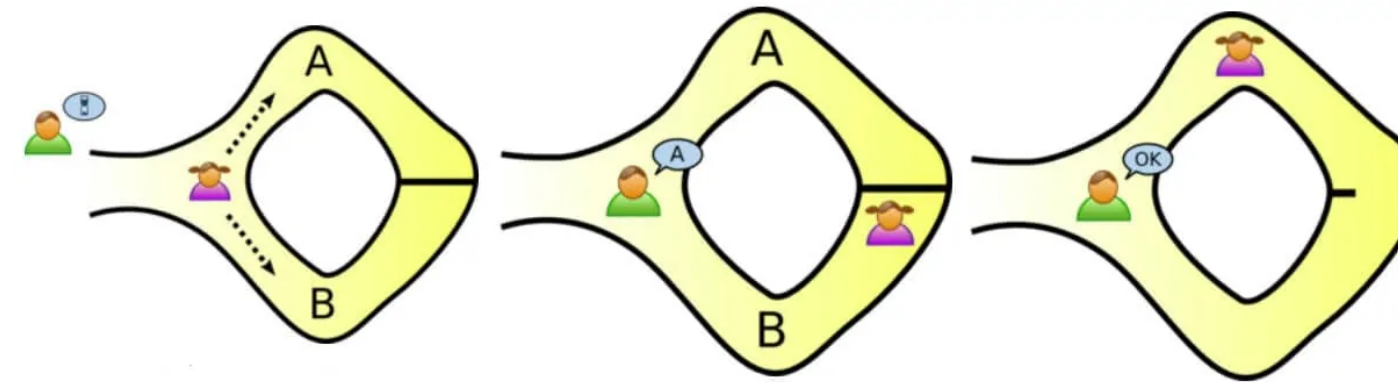
⁽³⁾ Anagram Laboratories, P.O. Box 791, Palo Alto CA 94301, USA.

The Strange Cave of Ali Baba

◇ Know, oh my children, that very long ago, in the Eastern city of Baghdad, there lived an old man named Ali Baba. Every day Ali Baba would go to the bazaar to buy or sell things. This is a story which is partly about Ali Baba, and partly also about a cave, a strange cave whose secret and wonder exist to this day. But I get ahead of myself ...

One day in the Baghdad bazaar a thief grabbed a purse from Ali Baba who right away started to run after him. The thief fled into a cave whose entryway forked into two dark winding passages: one to the left and the other to the right (The Entry of the Cave).

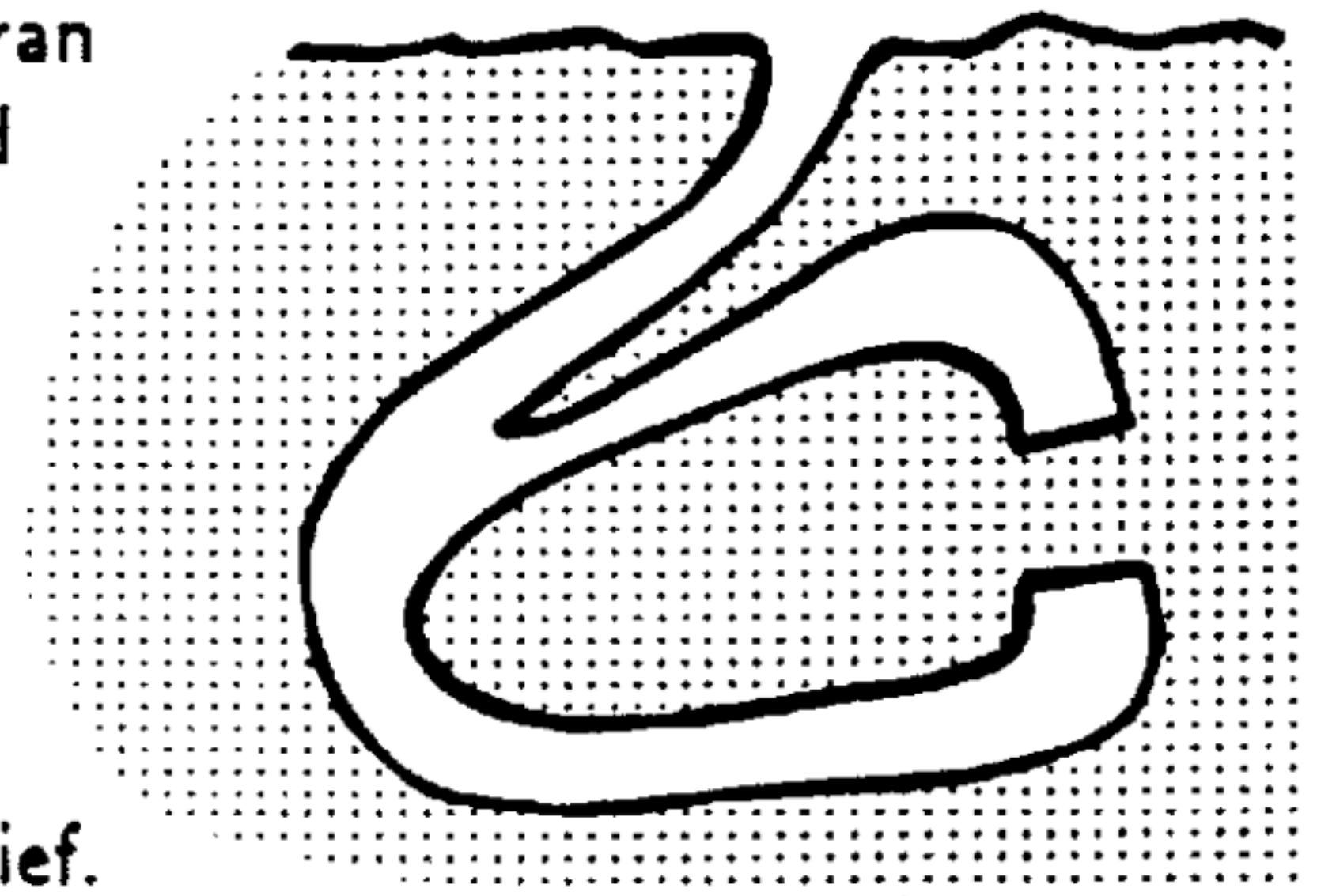
Zero-Knowledge Proof: Ali Baba Cave



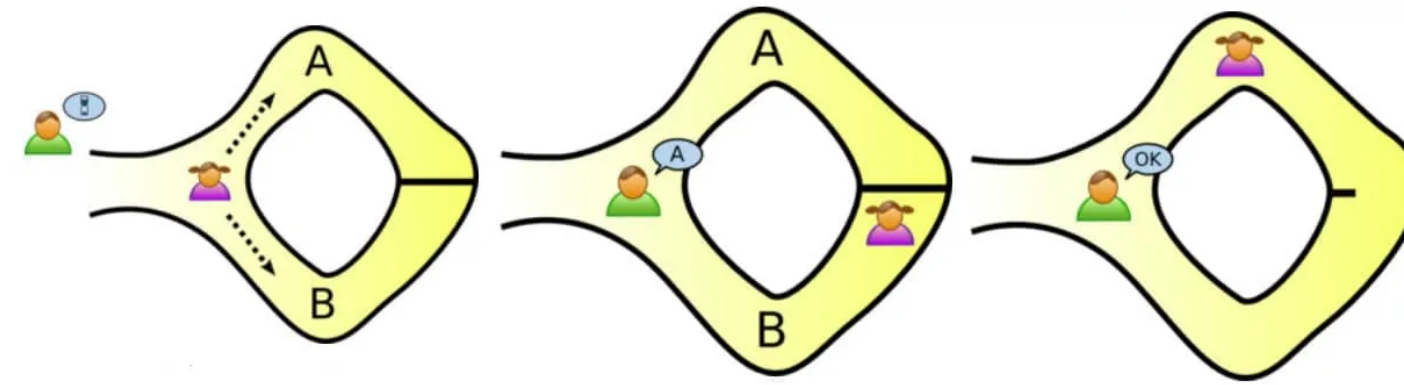
One day in the Baghdad bazaar a thief grabbed a purse from Ali Baba who right away started to run after him. The thief fled into a cave whose entryway forked into two dark winding passages: one to the left and the other to the right (The Entry of the Cave).

Ali Baba did not see which passage the thief ran into. Ali Baba had to choose which way to go, and he decided to go to the left. The left-hand passage ended in a dead end. Ali Baba searched all the way from the fork to the dead end, but he did not find the thief. Ali Baba said to himself that the thief was perhaps in the other passage. So he searched the right-hand passage, which also came to a dead end. But again he did not find the thief.

"This cave is pretty strange," said Ali Baba to himself, "Where has my thief gone?"



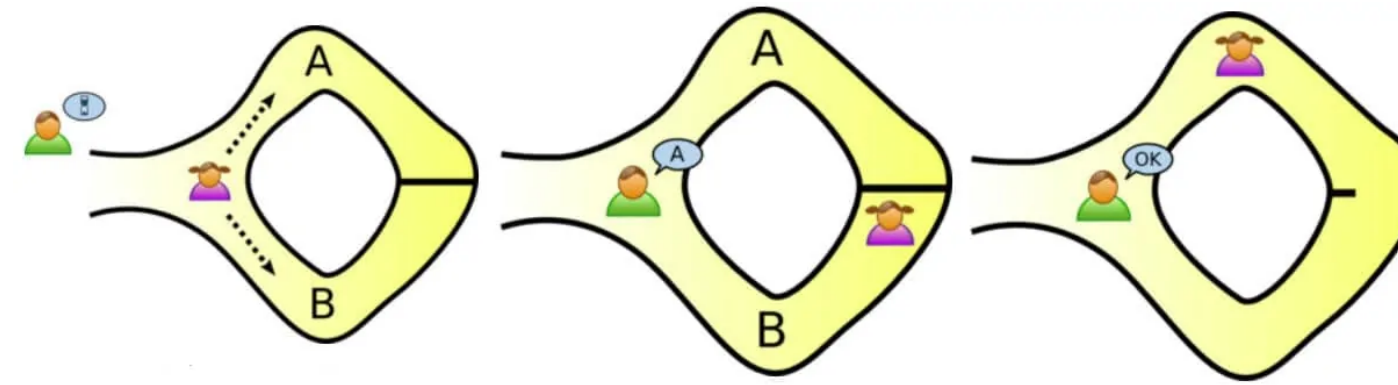
Zero-Knowledge Proof: Ali Baba Cave



The days went by, and every day brought its thief. Ali Baba always ran after the thief, but he never caught any of them. On the fortieth day a fortieth thief grabbed Ali Baba's turban and fled, as thirty-nine thieves had done before him, into the strange cave. Ali Baba yet again did not see which way the thief went. This time Ali Baba decided to search the left-hand passage, but again he did not find the thief at the end of the passage. Ali Baba was very puzzled.

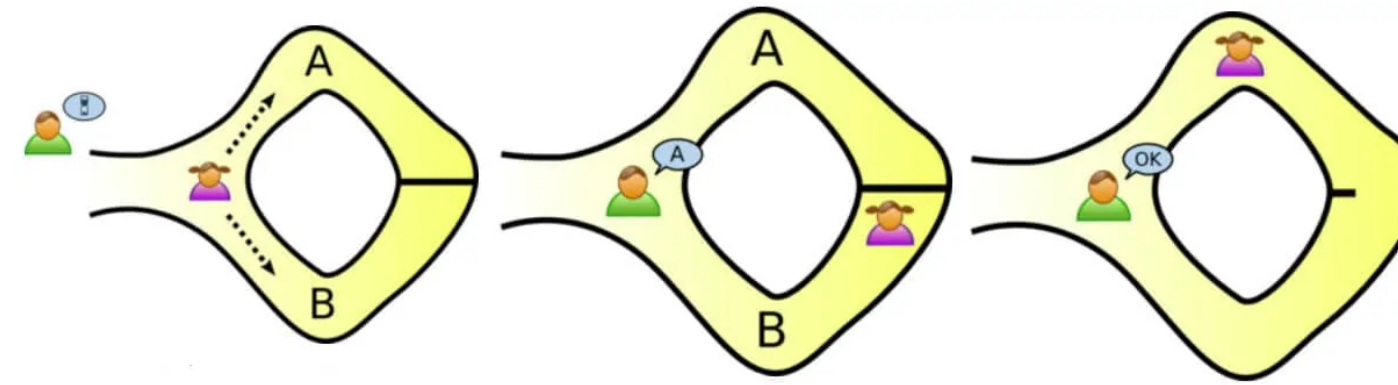
He could have said to himself, as he had done before, that the fortieth thief had been as lucky as each of the other thirty-nine thieves. But this explanation was so

Zero-Knowledge Proof: Ali Baba Cave



And Ali Baba set out to discover the secret of the strange cave. He decided to hide under some sacks at the end of the right-hand passage. After a very uncomfortable wait he saw a thief arrive who, sensing he was pursued by his victim, whispered the magic words, "Open sesame." Ali Baba was amazed to see the wall of the cave slide open. The thief ran through the opening. Then the wall slid closed again. The pursuer arrived, and was all upset to find only Ali Baba under the sacks at the dead end of the passage. The thief had escaped. But Ali Baba was all happy, for he was finding out the Secret of the Strange Cave.

Zero-Knowledge Proof: Ali Baba Cave

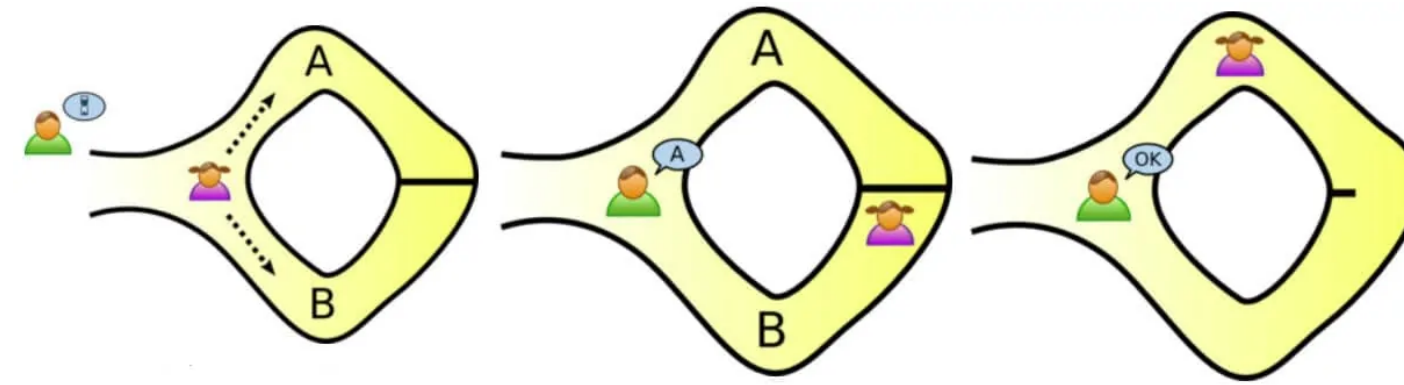


After several archaeological excavations in the ruins of the old Baghdad bazaar, the strange cave was located. It was not a myth! And, despite the centuries, the magic words still worked. All agog, the curious researchers went through the end wall between the two passages.

The television networks were quickly made aware of the unusual events taking place in Baghdad. A big American network even got an exclusive on the story. One of the researchers, a certain Mick Ali, a descendent perhaps of Ali Baba, wanted to demonstrate that he knew the secret. But he did not want to reveal the secret. Here is what he did.

First, a television crew filmed a detailed tour of the cave with the two dead-end passages. Then everybody went out of the cave. Mick Ali went back in alone and went down one of the passages. Then the reporter, accompanied by the camera, went inside only as far as the fork. There he flipped a coin to choose between right and left. If the coin came up heads he would tell Mick to come out on the right. If the coin came up tails he would tell Mick to come out on the left. It was heads, so the reporter called

Zero-Knowledge Proof: Ali Baba Cave

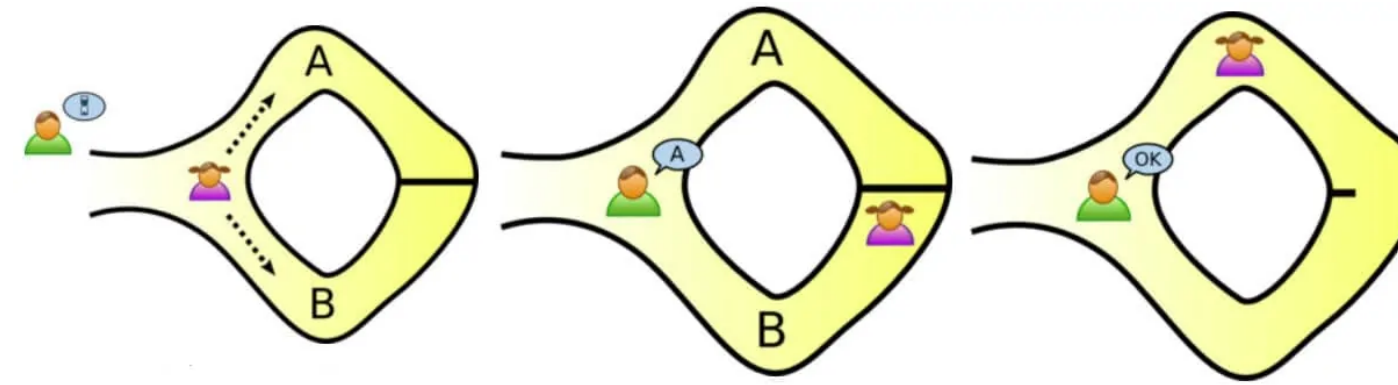


In memory of the forty thieves this demonstration scene was played forty times. Each of the times everybody went back out of the cave and Mick entered alone all the way in to one of the passages. Then the reporter and the camera went as far as the fork where he chose by flipping a coin which order to give to Mick. Mick succeeded in all forty scenes.

Employed by another television network, a jealous reporter wanted to also film a story on the strange cave. Mick refused to participate because he had given exclusive rights to the story to the first network.

But Mick mischievously suggested to the jealous reporter that the story could be filmed without possessing the secret. The jealous reporter thought and thought, and finally he understood. He said to himself, "I even know a stage actor who looks like Mick Ali and who could be mistaken for him."

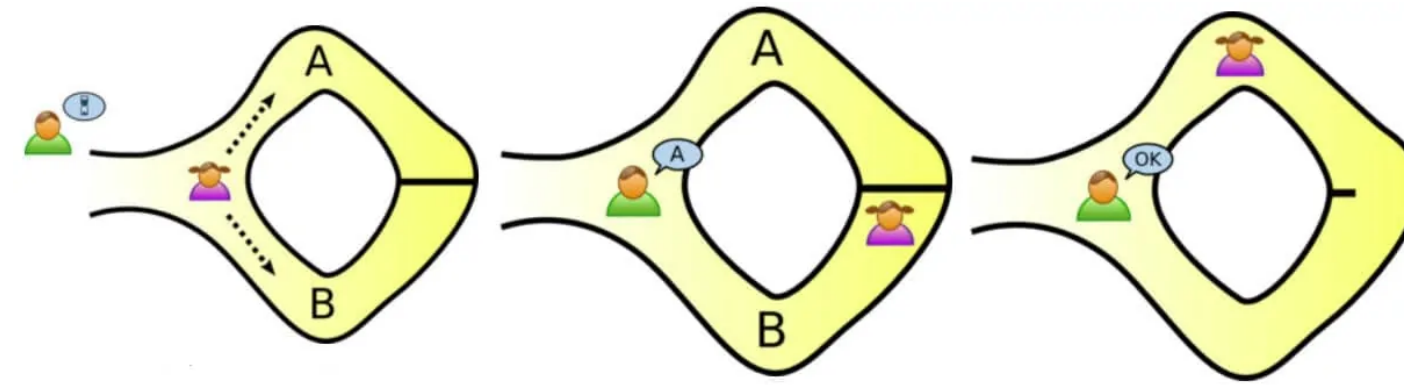
Zero-Knowledge Proof: Ali Baba Cave



And the second story was filmed. In the course of the filming half of the scenes were spoiled because Mick's double did not know how to get from one passage to the other! The jealous reporter edited the tape and only kept the successful scenes until he had forty of them.

The two stories were broadcast at the same hour on the same evening by the two competing American networks. The matter was taken to court. Both videotapes were placed into evidence. But the judges and the experts could not tell the tapes apart. Which tape was simulated? Which tape was genuine? The tapes alone were not enough to judge by.

Zero-Knowledge Proof: Ali Baba Cave



The simulation surely conveyed no knowledge of the secret. But the simulation and the genuine tape were indistinguishable. So the genuine tape did not convey knowledge of the secret either. The reporter who had gotten the exclusive story had been convinced at the time that Mick Ali knew the secret, but the reporter could not pass his conviction on to the judges in court or to the television audience either.

Mick Ali had achieved his real objective. He wanted, in fact, to show that it is possible to convince without revealing, and so without unveiling his secret.

Zero-Knowledge Proof

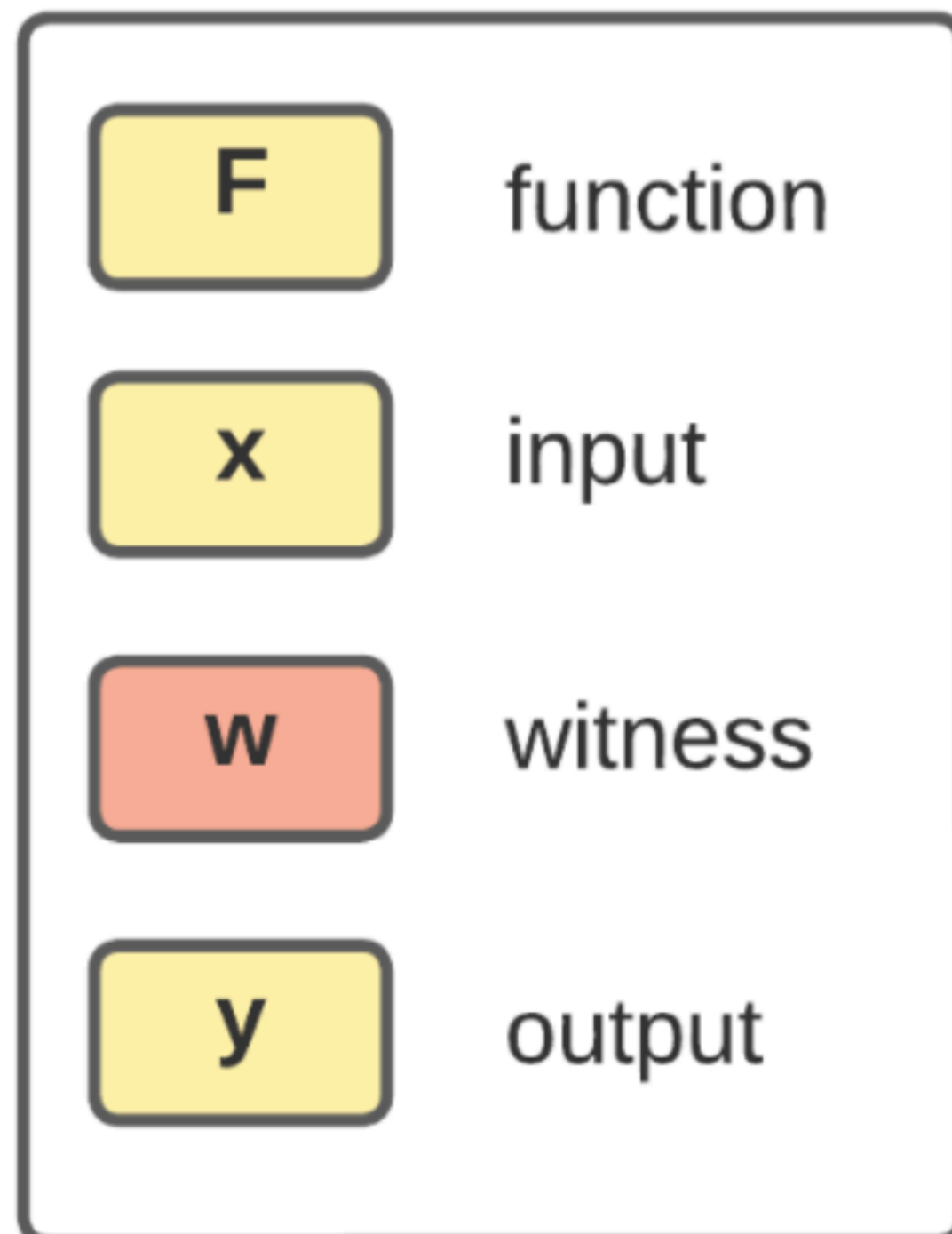
For a zero-knowledge protocol to be valid, it has to satisfy three properties:

- **Completeness:** If the proof is valid, it must be accepted by the verifier (ie. a correct proof is always accepted).
- **Soundness:** If the proof is invalid, it must be rejected by the verifier (ie. the probability that a verifier accepts an incorrect solution is negligible).
- **Zero-Knowledge:** The verifier does not learn any information about the information that the prover claims to know other than the assertion that she knows it.

Zero-Knowledge Proof

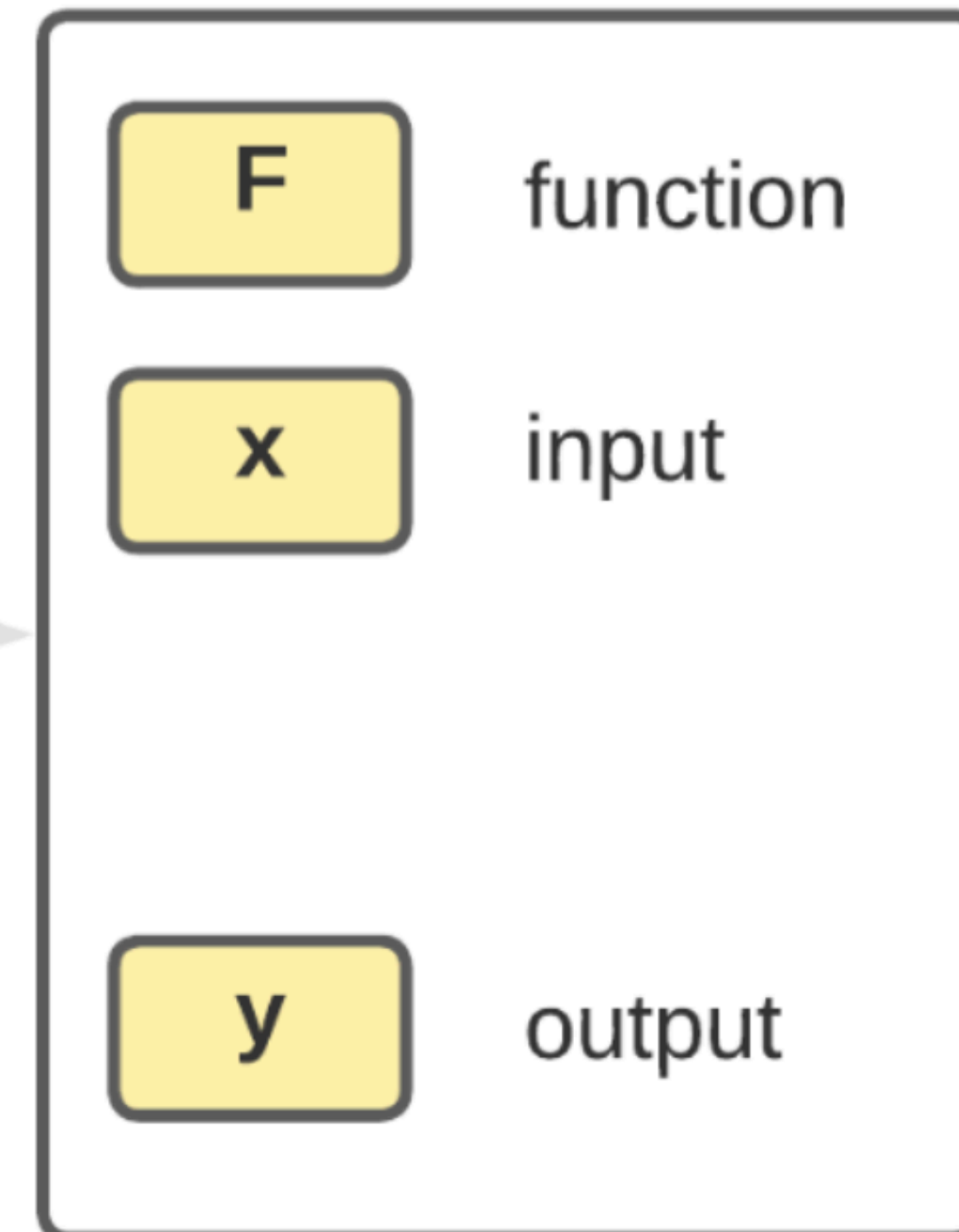
I know w such that $F(x, w) = y$

Prover



Proof

Verifier



Zero-Knowledge Proof: Formal Definition

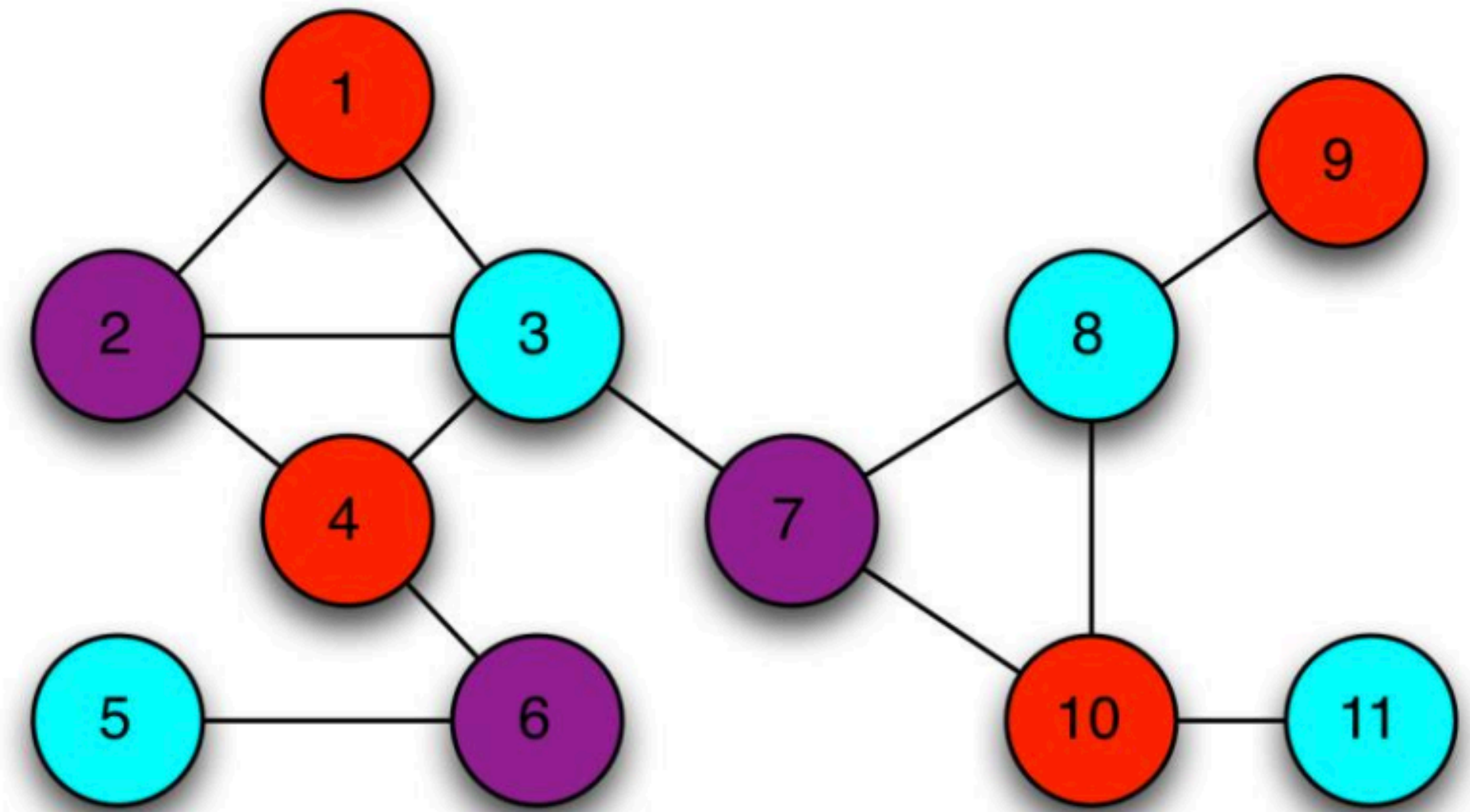
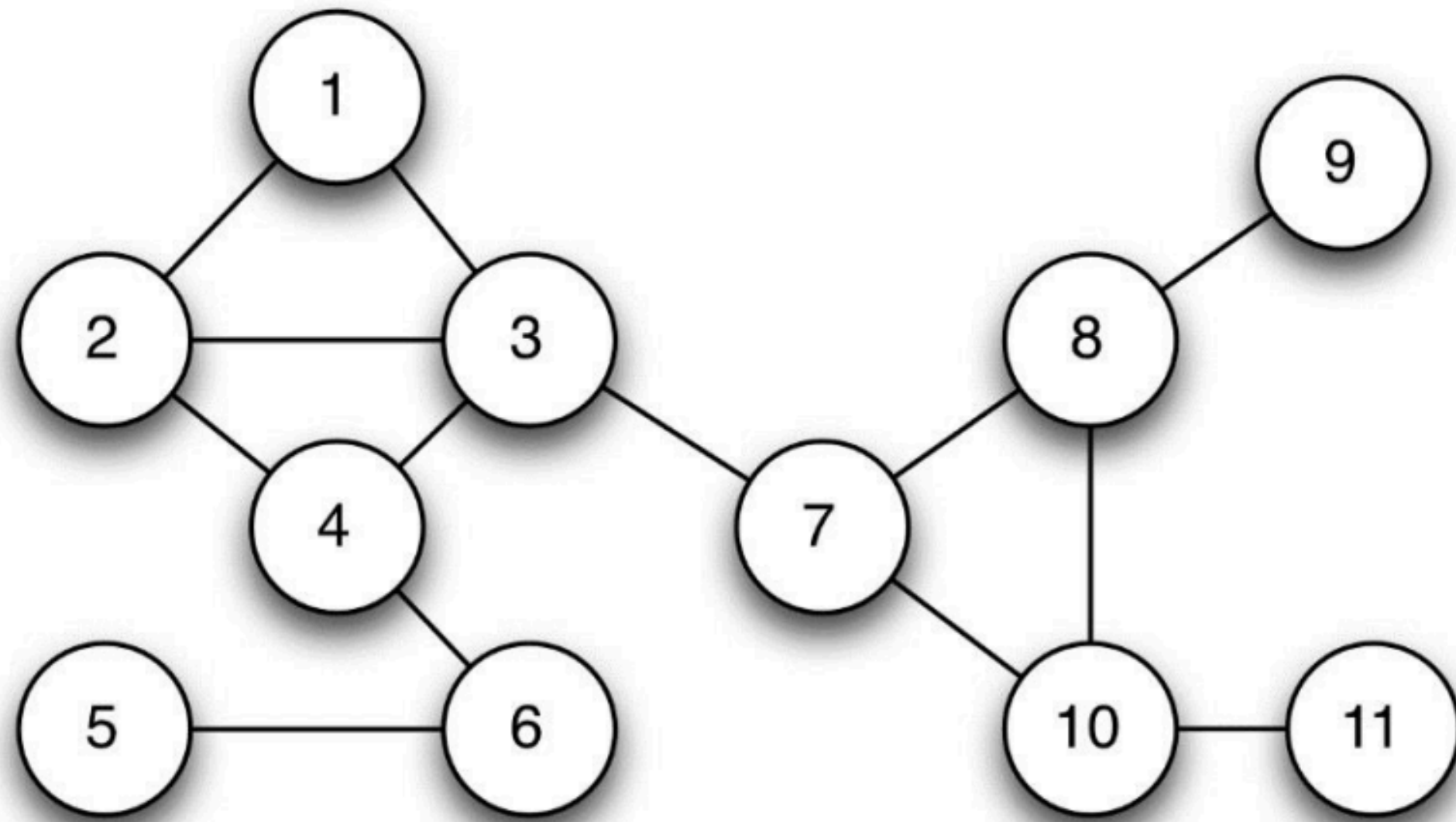
An IBS is a protocol (P, V) between a Prover and a Verifier that decides $L \in \Sigma^*$ if $\forall x \in \Sigma^*, |x| = n$

1. If $x \in L$ then P accepts with $P(x \in L) \geq (1 - 2^{-n})$
2. If $x \notin L$, there is no IBS such as P accepts with $P(x \in L) > 2^{-n}$

Such an IBS is zero knowledge for L , if for any probabilistic polynomial time (PPT) verifier $\hat{V}$ there exists a PPT simulator S such that

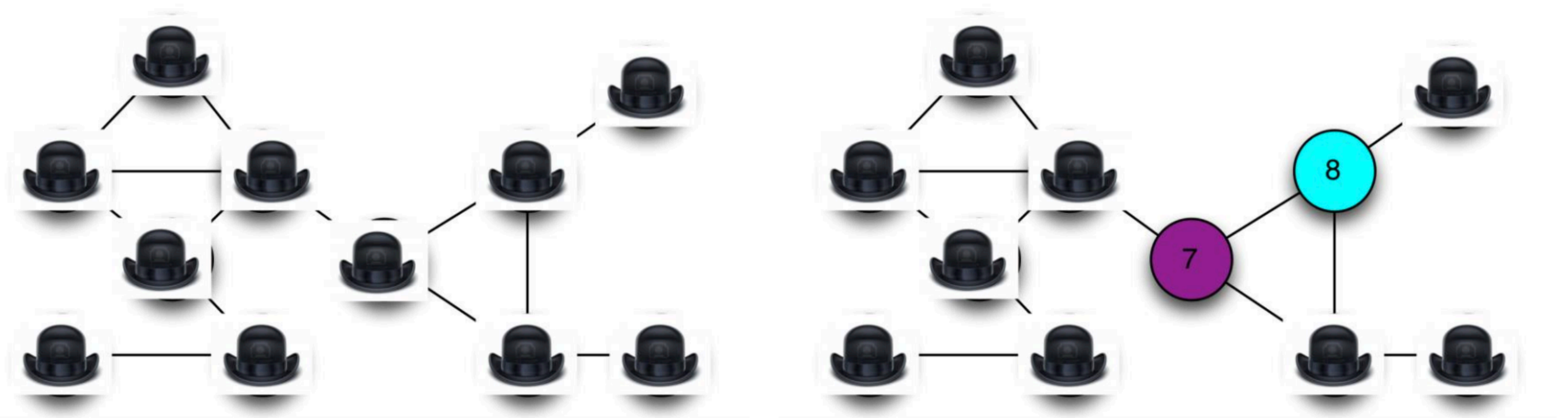
$$\forall x \in L, z \in \{0,1\}^*, \text{View}_{\hat{V}} [P(x) \leftrightarrow \hat{V}(x,z)] = S(x,z)$$

Zero-Knowledge Proof: Graph 3-Coloring



complexity class: NP-Complete

Zero-Knowledge Proof: Graph 3-Coloring



Propability of succesful cheat: $<(1-1/E)$

Zero-Knowledge Proof: SNARKs

In order to explain the solution to this problem, we need to understand the basic elements of zero-knowledge proofs — specifically “zk-SNARK” proofs.

zk-SNARK is an acronym that stands for “Zero-Knowledge Succinct Non-Interactive Argument of Knowledge.” Let’s break that down:

- **ZK = Zero knowledge:** We can prove a fact without revealing that fact.
- **S = Succinct:** The proofs for even very large programs are small (eg. a few hundred bytes) and can be verified within a few milliseconds.
- **N = Non-interactive:** There is no need for back-and-forth communication between a prover and verifier (eg. such as in the “Color Blindness” example above). Instead, a prover simply hands over a proof to the verifier and the verifier can compute whether the proof is valid or invalid.
- **AR = Argument:** For technical reasons, the proofs are called “arguments” because they are not technically “proofs” in the traditional sense — but they functionally serve the same purpose. The difference between “proof” and “argument” is very technical and out of scope for this post.
- **K = Knowledge:** This refers to the information that the prover knows.

Zero-Knowledge Proof: Setup

Setup

The Setup algorithm generates two keys: the proving key (“pk”) and verification key (“vk”). In order to generate these keys, it takes as inputs a secret parameter (“lambda”) and a program C (such as the one we defined above, which returns true if the hash of “w” matches “x”).

$$(pk, vk) = Setup(lambda, C)$$

Zero-Knowledge Proof: Prove

Prove

The prove algorithm is responsible for generating the zero-knowledge proof. It takes as inputs the proving key (“pk”), the private witness “w,” and a public value “x” to generate the proof.

$$proof = Prove(pk, x, w)$$

Zero-Knowledge Proof: Verify

Verify

The verify algorithm is responsible for taking the proof created by the proving algorithm and asserting true or false, depending on whether the proof is correct or incorrect. It takes as inputs the verification key (“vk”), the public value “x,” and the “proof”.

$$\textit{Verify}(vk, x, \textit{proof}) = \textit{bool}$$

Zero-Knowledge Proof: Variants

Variants of zero-knowledge [\[edit \]](#)

Different variants of zero-knowledge can be defined by formalizing the intuitive concept of what is meant by the output of the simulator "looking like" the execution of the real proof protocol in the following ways:

- We speak of *perfect zero-knowledge* if the distributions produced by the simulator and the proof protocol are distributed exactly the same. This is for instance the case in the first example above.
- *Statistical zero-knowledge*^[13] means that the distributions are not necessarily exactly the same, but they are **statistically close**, meaning that their statistical difference is a **negligible function**.
- We speak of *computational zero-knowledge* if no efficient algorithm can distinguish the two distributions.

ZoKrates

ZoKrates is a toolbox for zkSNARKs on Ethereum. It helps you use verifiable computation in your DApp, from the specification of your program in a high level language to generating proofs of computation to verifying those proofs in Solidity.

One particular family of ZKPs is described as zero-knowledge **S**uccinct **N**on-interactive **AR**guments of **K**nowledge, a.k.a. zkSNARKs. zkSNARKs are the most widely used zero-knowledge protocols, with the anonymous cryptocurrency Zcash and the smart-contract platform Ethereum among the notable early adopters.

ZoKrates

Ethereum runs computations on all nodes of the network, resulting in high costs, limits in complexity, and low privacy. zkSNARKs have been enabling to only verify computations on-chain for a fraction of the cost of running them, but are hard to grasp and work with.

ZoKrates bridges this gap. It helps you create off-chain programs and link them to the Ethereum blockchain, expanding the possibilities for your DApp.

We provide one-line installation for Linux, MacOS and FreeBSD:

```
curl -LSfs get.zokrat.es | sh
```